

Received 6 October 2024; revised 3 May 2025; accepted 6 May 2025; date of publication 12 May 2025; date of current version 30 May 2025.

Digital Object Identifier 10.1109/TQE.2025.3568865

Three-Party Controlled Authentication Semiquantum Key Agreement Protocol for Online Joint Consultation

XIYUAN LIANG^{ID}, YEFENG HE^{ID}, YICHI ZHANG, AND JIAQIANG FAN

Faculty of Cyberspace Security, Xi'an University of Posts and Telecommunications, Xi'an 710100, China

Corresponding author: YeFeng He (e-mail: yefenghe2008@hotmail.com).

This work was supported in part by the National Natural Science Foundation of China under Grant 61802302 and in part by the Basic Research Project of Natural Science of Shaanxi Province under Grant 2021JM-462.

ABSTRACT In Wise Information Technology of Medicine, to ensure both confidentiality and integrity of the data created during online joint consultations, and to solve the problem that ordinary users cannot afford expensive quantum devices and are vulnerable to man-in-the-middle attacks during communication, this article proposes a three-party controlled authentication semiquantum key agreement protocol, leveraging the measurement retransmission operation and the entanglement properties of cluster states. With the help of a trusted controller with full quantum capabilities, the identities of three semiquantum parties are authenticated, and a shared key is negotiated fairly for subsequent communication. Since the semiquantum participants only need to perform simple quantum state preparation, measurement, and reflection operations, the protocol reduces the requirements for participants' capabilities and equipment. Moreover, the protocol prevents man-in-the-middle attacks by authenticating the identity of participants. The security evaluation demonstrates that the protocol is capable of effectively defending against both internal participant threats and external intrusions. Moreover, a comparison with existing semiquantum key agreement protocols reveals that this protocol offers certain advantages when its functionality and performance are comprehensively evaluated.

INDEX TERMS Authorizing identity, cluster states, quantum cryptography, qubit efficiency, semiquantum key agreement (SQKA).

I. INTRODUCTION

Wise Information Technology of Medicine (WITMED) uses advanced information technologies to provide patients with personalized diagnosis and treatment, telemedicine, disease data management, and analysis services to improve the efficiency and quality of medical services. Taking patients with malignant tumors as an example, the patient's condition often involves multiple organs, multiple systems, and multiple disciplines. When seeking medical treatment, they not only need to be transferred to multiple departments in different hospitals, but also often undergo repeated examinations and excessive treatment, which may cause them to miss the best treatment opportunity. WITMED allows such patients to receive remote cross-hospital multidisciplinary joint consultations, with a team of authoritative experts and professors from various departments. During the consultation period, the patient's condition data and the diagnosis and treatment plan generated after the expert discussion need to be

encrypted to protect the patient's privacy, while the integrity of data also needs to be guaranteed to prevent data from being tampered with.

Different from the traditional cryptography that relies on mathematically difficult problems to ensure security, the security of quantum cryptography is based on the basic principles of quantum mechanics, so it is unconditionally secure in theory [1]. To encrypt data and ensure data integrity during communication, the key lies in quantum key distribution (QKD) [1], [2] and quantum key agreement (QKA) [3], [4]. Unlike QKD protocols, where one participant selects and distributes the session key, QKA allows all participants to jointly negotiate the session key, with each contributing equally to its generation. Therefore, it is necessary to use the QKA protocol to negotiate the session key to protect data privacy and integrity under WITMED, especially in remote multidisciplinary joint consultation.

Since Zhou et al. [3] proposed the first QKA protocol, many different QKA protocols have appeared. Shen et al. [4] proposed an efficient QKA protocol based on the four-particle cluster states, but the protocol only allows two participants to communicate. After that, many multiparty QKA protocols were proposed. Shi and Zhong [5] first extended the two-party QKA protocol to multiparty and proposed a multiparty QKA protocol based on Bell states and Bell measurements. However, Liu et al. [6] found that Shi's protocol is unfair: dishonest participants can independently determine the key. At the same time, they proposed a secure multiparty QKA protocol based on single particles. He and Ma [7] proposed two- and three-party QKA protocols based on unitary operations and four-particle cluster states, respectively. The protocol proposed by Sun et al. [8] extended Shen's two parties to multiparty cases. The protocol allowed N participants to share a key. Yang et al. [9] proposed an efficient tripartite QKA protocol and extended it to the N party. The scheme replaces the N -particle entangled states with the three-particle entangled states in the case of extending to the N party, which saves quantum resources. Liu et al. [10] designed a two-layer multiparty QKA protocol based on non-maximally entangled states and proposed a new method to solve the key generation problem. Since the security of the QKA protocol is also affected by the channel noise, some researchers have also spent energy studying the antinoise QKA protocols [11], [12], [13]. These protocols can resist collective-dephasing noise and collective-rotation noise.

To adapt to different application environments, researchers have also proposed a controlled QKA protocol. For example, Tang et al. [14] proposed a two- and three-party controlled QKA protocol based on three-particle greenberger-horne-zeilinger (GHZ) states and Bell measurements. Compared with the previous QKA protocol, the protocol introduces a monitor to control the protocol process to improve the controllability of the protocol. Arindam and Anirban [15] proposed a new controlled QKA protocol based on Bell states and single-photon states. Compared with Tang's scheme, the scheme does not require quantum storage, and the quantum state used is easier to generate and maintain than the GHZ state.

In addition, the traditional QKA protocol cannot authenticate the identity of the participants and is vulnerable to man-in-the-middle attacks. Therefore, the mutual authentication QKA protocol is also an important research direction for scholars. Zhu et al. [16] proposed a semihonest tripartite mutual authentication QKA protocol based on GHZ-like states. The protocol introduces a third party to provide entangled particles for authentication and key agreement. Ma et al. [17] proposed a quantum teleportation scheme using local operation and five-particle cluster states, which realized mutual authentication and key agreement between two clients in different fields. He et al. [18] proposed a two-way authentication QKA protocol based on Bell states. The protocol uses the measurement correlation of Bell states to authenticate

each other's identity and uses the entanglement exchange relationship of Bell states to negotiate the session key fairly. He et al. [19] proposed a mutual authentication QKA protocol based on single-particle states and four-particle GHZ states. The protocol uses the hash value and random number of shared identity information for identity authentication and uses the measurement correlation characteristics of the four-particle GHZ states to negotiate the shared key. Xu et al. [20] proposed a new tripartite QKA protocol based on the GHZ entangled state. In the key agreement phase, each participant obtains the bitwise XOR of the remaining participant's key values by measuring the particles after the Pauli operations, rather than extracting their private keys. Wang et al. [21] proposed a two-party secure mutual authentication QKA scheme with key recovery based on single-photon states and Bell states, which realized mutual authentication and minimized quantum key consumption.

However, the aforementioned protocols assume that all participants possess full quantum capabilities. In practical applications, due to the high costs and limited resources, most participants cannot afford high-performance quantum devices. Therefore, simplifying the quantum operations required by participants is essential for easier protocol implementation. To address this issue, Shukla et al. [22] proposed a two-party semiquantum key agreement (SQKA) protocol based on Bell states. In the SQKA protocol [23], [24], [25], [26], one participant possesses full quantum capabilities, while the others have only semiquantum capabilities. Semiquantum participants are limited to performing two operations: 1) reflection operation (CTRL), which returns the received particles directly without any operation on the received particles and 2) measurement operation (SIFT), in which the semiquantum participants only perform Z-based measurements on the received particles and generate new particles according to the measurement results.

After that, researchers used more different quantum states in the design of SQKA. He et al. [23] proposed a two-party SQKA protocol based on G-like states, which enables two semiquantum participants to fairly establish a secure shared key with the assistance of a fully quantum-capable trusted third party. Xu et al. [24] proposed a semiquantum multiparty QKA protocol based on multiparticle GHZ entangled states. The protocol does not require preshared keys between participants, nor does it require unitary operation or quantum entanglement swapping to achieve key agreement, which reduces the requirements for participants' capabilities and devices. He et al. [25] proposed a four-party SQKA protocol based on four-particle cluster states. The protocol allows one fully quantum party and three semiquantum parties to fairly negotiate a shared key without relying on a third party, thus increasing the number of participants. However, the quantum efficiency of this protocol is relatively low compared to other protocols. Yi et al. [26] proposed an SQKA protocol based on W states, which uses four different form of W states. However, the previously proposed protocols do not take

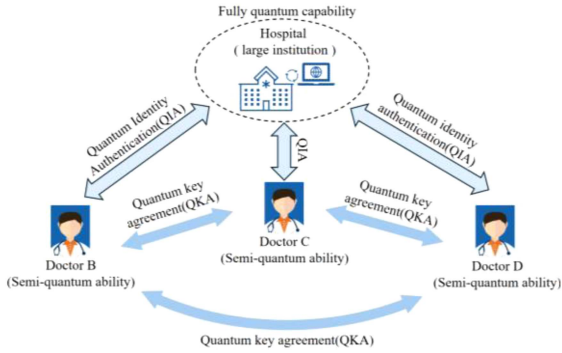


FIGURE 1. CASQKA model.

into account that the SQKA protocol is also vulnerable to man-in-the-middle attacks, which poses significant security risks.

In practical applications, different scenarios have different requirements for the functions of the QKA protocol. Taking the online joint consultation of doctors to patients in the WITMED scenario as an example, to facilitate the broader adoption of joint consultations, the designed protocol should reduce the quantum equipment requirements for participating doctors. Simultaneously, to ensure the security of communications and prevent the risk of impersonation, identity authentication for participating doctors is necessary. Therefore, the protocol should incorporate both semiquantum functionality and identity authentication as dual safeguards.

In this article, we propose a three-party controlled authentication semiquantum key agreement (CASQKA) protocol based on the cluster state. With the help of the trusted controller Alice, who has full quantum capabilities (representing hospitals or large institutions with strong quantum devices), the semiquantum participants Bob, Charlie, and Dave (representing doctors participating in the joint consultation) complete identity authentication and conduct fair key agreement. In addition, we prove the correctness of the proposed scheme, which can effectively resist participant attacks and external attacks and also has advantages in performance.

II. NEW THREE-PARTY CASQKA PROTOCOL

A. PROTOCOL MODEL

As shown in Fig. 1, the hospital (large institution) is a trusted controller with fully quantum capabilities. It is responsible for authenticating the identities of doctors B, C, and D participating in multidisciplinary joint consultation to ensure the correctness of the identities of the consultants, and for providing assistance when doctors B, C, and D negotiate shared keys. Doctors B, C, and D will negotiate a shared key to ensure the confidentiality and integrity of the data generated when discussing the patient's condition and customizing the diagnosis and treatment plan.

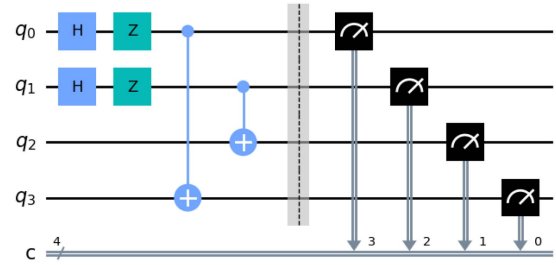


FIGURE 2. Quantum circuit diagrams of preparing and measuring cluster states.

Due to the restrictive structure of the quantum state selected in this article, its quantum entanglement characteristics only act on four particles, so the protocol supports up to four participants (including the hospital and three doctors) to communicate.

B. BASIC KNOWLEDGE

The cluster state, put forward by Briegel and Raussendorf [27], is a quantum state that displays unique properties when the number of particles exceeds 3. The cluster state has maximum connectivity, better persistent entanglement than the GHZ state, and can resist qubit loss. The four-particle cluster state can be represented as

$$\begin{aligned} |\Phi\rangle_{abcd} &= \frac{1}{2} (|0000\rangle + |0110\rangle + |1001\rangle + |1111\rangle) \\ &= \frac{1}{\sqrt{2}} (|0\rangle_a |\varphi^+\rangle_{bc} |0\rangle_d + |1\rangle_a |\varphi^-\rangle_{bc} |1\rangle_d) \\ &= \frac{1}{\sqrt{2}} (|0\rangle_b |\varphi^+\rangle_{ad} |0\rangle_c + |1\rangle_b |\varphi^-\rangle_{ad} |1\rangle_c). \end{aligned} \quad (1)$$

As shown in Fig. 2, the preparation and measurement of cluster states can be realized by quantum circuits [28].

C. NEW THREE-PARTY CASQKA PROTOCOL

Assume that semiquantum doctors B, C, and D wish to negotiate a key together. They need to complete the identity authentication with the assistance of the hospital (large institution), which is a trusted controller with full quantum capabilities, and then, they negotiate a shared key. For convenience, the hospital (large institution) is referred to as Alice, and doctors B, C, and D are referred to as Bob, Charlie, and Dave, respectively.

Bob, Charlie, and Dave secretly share the identity information sequences $R_B = \{R_B^1, R_B^2, \dots, R_B^n\}$, $R_C = \{R_C^1, R_C^2, \dots, R_C^n\}$, $R_D = \{R_D^1, R_D^2, \dots, R_D^n\}$ with Alice in advance, where $R_B^i, R_C^i, R_D^i \in \{0, 1\}$ and $i \in \{1, 2, \dots, n\}$. Before the beginning of the protocol, Bob, Charlie, and Dave randomly generate their own private keys $K_B = \{K_B^1, K_B^2, \dots, K_B^{4n}\}$, $K_C = \{K_C^1, K_C^2, \dots, K_C^{4n}\}$, and $K_D = \{K_D^1, K_D^2, \dots, K_D^{4n}\}$, where $K_B^i, K_C^i, K_D^i \in \{0, 1\}$ and $i \in \{1, 2, \dots, 4n\}$. The specific protocol flow is shown in Fig. 3.

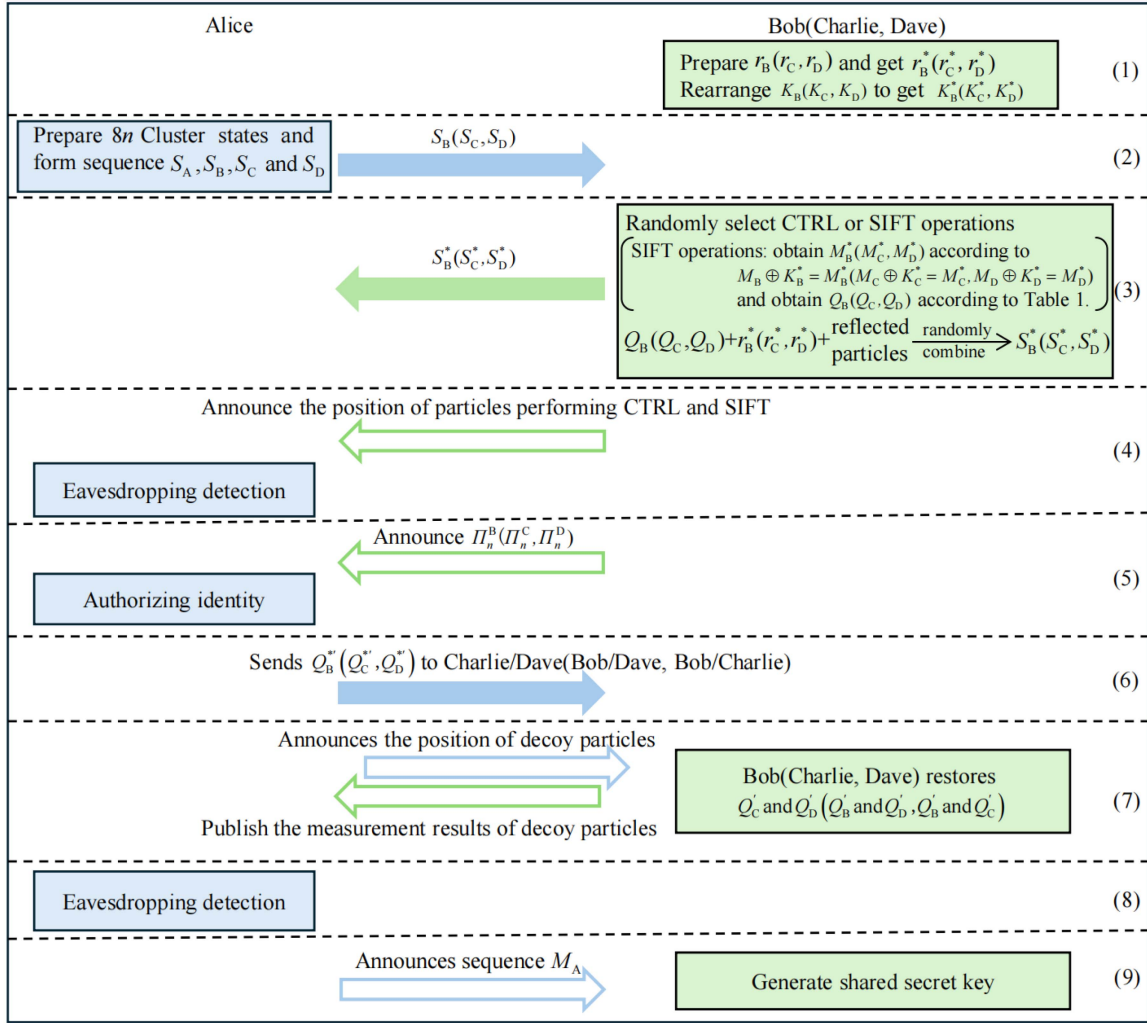


FIGURE 3. Workflow of the proposed protocol.

TABLE 1. Coding Rules

Binary information	Particle states
0	$ 0\rangle$
1	$ 1\rangle$

1) PREPARATION STAGE

(1) Bob, Charlie, and Dave prepare quantum sequences $r_B = r_B^1 r_B^2, \dots, r_B^n$, $r_C = r_C^1 r_C^2, \dots, r_C^n$, and $r_D = r_D^1 r_D^2, \dots, r_D^n$ corresponding to their own identity information sequences R_B, R_C , and R_D , and the preparation rules are shown in Table 1. Then, Bob (Charlie and Dave) chooses the permutation operator $\Pi_n^B(\Pi_n^C, \Pi_n^D)$ to rearrange $r_B(r_C, r_D)$ to get $r_B^*(r_C^*, r_D^*)$ and chooses the permutation operator $\Pi_{4n}^B(\Pi_{4n}^C, \Pi_{4n}^D)$ to rearrange $K_B(K_C, K_D)$ to get $K_B^*(K_C^*, K_D^*)$.

2) PREPARATION AND TRANSMISSION OF CLUSTER STATES

(2) Alice prepares $8n$ four-particle cluster state $|\Phi\rangle_{ABCD}$ to form a cluster state sequence. Next, Alice divides the

cluster state sequence into S_A, S_B, S_C , and S_D , where $S_T (T = A, B, C, D)$ is composed of all T particles in the cluster state sequence. Finally, Alice sends the sequences S_B, S_C , and S_D to Bob, Charlie, and Dave, respectively, through the quantum channel and retains S_A by herself.

3) EAVESDROPPING DETECTION STAGE

(3) After Bob, Charlie, and Dave receive S_B, S_C , and S_D , they perform the following operations on the particles in the sequence.

- I) $4n$ particles are randomly selected to perform the reflection operation (CTRL operation).
- II) Z-based measurements are performed on the remaining $4n$ particles, and the resulting measurements are decoded into binary sequences M_B, M_C , and M_D according to Table 1. The new quantum state sequences Q_B, Q_C , and Q_D are prepared according to the following rules (SIFT operation): Bob uses formula $M_B \oplus K_B^* = M_B^*$ to obtain the binary sequence $M_B^* =$

TABLE 2. Operations of Alice in Eight Situations

Case	Bob's choice	Charlie's choice	Dave's choice	Alice's action
1	CTRL	CTRL	CTRL	Cluster joint measurement is performed on the particles of Alice, Bob, Charlie and Dave.
2	CTRL	CTRL	SIFT	Bell measurement is performed on Bob and Charlie's particles
3	CTRL	SIFT	CTRL	Bell measurement is performed on Alice and Dave's particles
4	CTRL	SIFT	SIFT	Z-based measurement is performed on Bob's particles. Alice is informed by Charlie and Dave of the measurement results of M_C^i and M_D^i
5	SIFT	CTRL	CTRL	Bell measurement is performed on Alice and Dave's particles
6	SIFT	CTRL	SIFT	Z-based measurement is performed on Charlie's particles. Alice is informed by Bob and Dave of the measurement results of M_B^i and M_D^i
7	SIFT	SIFT	CTRL	Z-based measurement is performed on Dave's particles. Alice is informed by Bob and Charlie of the measurement results of M_B^i and M_C^i
8	SIFT	SIFT	SIFT	Retains particles of Bob, Charlie, and Dave

$\{M_B^{1*}, M_B^{2*}, \dots, M_B^{4n*}\}$. Then, according to Table 1, the binary sequence Q_B^* is encoded as a quantum state sequence $Q_B = \{Q_B^1, Q_B^2, \dots, Q_B^{4n}\}$. Similarly, Charlie and Dave obtained the binary sequences M_C^* and M_D^* and the new quantum state sequences Q_C and Q_D , respectively.

Bob (Charlie and Dave) randomly combines the particles in the sequences $Q_B(Q_C, Q_D)$ and $r_B^*(r_C^*, r_D^*)$ and the reflected particles in (I) to obtain a new sequence $S_B^*(S_C^*, S_D^*)$ and then sends the sequence $S_B^*(S_C^*, S_D^*)$ to Alice.

(4) After Alice receives S_B^* , S_C^* , and S_D^* , Bob, Charlie, and Dave announce the location of particles performing CTRL and SIFT operations through the classical authentication channel. According to the information published by Bob, Charlie, and Dave, Alice can restore the sequences r_B^* , r_C^* , and r_D^* and perform Z-based measurement on particles in r_B^* , r_C^* , and r_D^* . Then, decode the measurement results into binary sequences R_B^* , R_C^* , and R_D^* according to Table 1. Subsequently, based on the choices of Bob, Charlie, and Dave, Alice performs different operations on the received particles. There are eight cases, as shown in Table 2.

In Cases 1–7, Alice determines the error rate by comparing the measurement results with (1). If the error rate falls below the threshold, the protocol proceeds; if not, the agreement terminates and returns to Step (1). In Case 8, Bob, Charlie, and Dave all perform SIFT operations. At this time, Alice retains the particles of Bob, Charlie, and Dave to form sequences Q'_B , Q'_C , and Q'_D , respectively.

4) IDENTITY AUTHENTICATION STAGE

(5) After the eavesdropping detection is passed, Bob, Charlie, and Dave publish permutation operators Π_n^B , Π_n^C , and Π_n^D through secure classical channels. Alice can get $R'_B(R'_C, R'_D)$ by permutation operator $\Pi_n^B(\Pi_n^C, \Pi_n^D)$ and binary sequence $R_B^{\prime\prime}(R_C^{\prime\prime}, R_D^{\prime\prime})$. Then, by comparing whether the values of $R_B^{\prime\prime}(R_C^{\prime\prime}, R_D^{\prime\prime})$ and $R_B(R_C, R_D)$ are equal, the identity of Bob (Charlie and Dave) is verified. If the error rate falls below the preagreed threshold, the identity of Bob (Charlie and Dave)

is successfully authenticated. If Bob, Charlie, and Dave all pass the identity authentication, the protocol advances to the next step, alternatively, it terminates.

5) KEY AGREEMENT STAGE

(6) Once the identity authentication is successfully completed, Alice randomly inserts the particles reflected by Bob, Charlie, and Dave in Case 1 as decoy states into the sequences Q'_B , Q'_C , and Q'_D and obtains the sequences $Q_{B'}^{*}$, $Q_{C'}^{*}$, and $Q_{D'}^{*}$. Then, Alice sends sequence $Q_{B'}^{*}$ to Charlie and Dave, $Q_{C'}^{*}$ to Bob and Dave, and $Q_{D'}^{*}$ to Bob and Charlie.

(7) After Bob (Charlie and Dave) receives the sequences $Q_C^{*/}$ and $Q_D^{*/}$ ($Q_B^{*/}$ and $Q_D^{*/}$, $Q_B^{*/}$ and $Q_C^{*/}$), Alice informs Bob, Charlie, and Dave of the position of the decoy particles inserted through the secure classical channel. Subsequently, Bob (Charlie and Dave) restores Q'_C and Q'_D (Q'_C and Q'_D , Q'_B and Q'_C) and performs Z-based measurement on the decoy particles and then publishes the measurement results. Simultaneously, Alice performs Z-based measurements on the particles at the corresponding positions in her hand.

(8) Alice calculates the error rate according to (1) with the results measured by herself and the results published by Bob, Charlie, and Dave. If the error rate falls below the threshold, the quantum channel is considered secure, and the protocol can continue. Alternatively, the agreement terminates and returns to Step (1).

(9) Alice performs Z-based measurement on the sequence S_A , decodes it to obtain the binary sequence M_A (according to Table 1), and publishes it. At the same time, Bob, Charlie, and Dave publish the permutation operators Π_{4n}^B , Π_{4n}^C , and Π_{4n}^D . Bob (Charlie and Dave) performs Z-based measurements on Q'_C and Q'_D (Q'_B and Q'_D , Q'_B and Q'_C) and decodes the binary sequences $M_C^{*'} and $M_D^{*'} (M_B^{*'} and $M_D^{*'} (M_B^{*'} and $M_C^{*'})$. Then, according to (1), Bob can get the value of $K'_C \oplus K'_D$ by using M_A and M_B and permutation operators Π_{4n}^C and Π_{4n}^D . Similarly, Charlie can get the value of $K'_B \oplus K'_D$; Dave can get the value of $K'_B \oplus K'_C$, where $K'_B = \{K_B^1, K_B^2, \dots, K_B^{n'}\}$, $K'_C = \{K_C^1, K_C^2, \dots, K_C^{n'}\}$, and $K'_D = \{K_D^1, K_D^2, \dots, K_D^{n'}\}$.$$$

TABLE 3. Process of Performing SIFT Operation

M_B	K'_B	$M_B^* = M_B \oplus K'_B$	Q_B	M_C	K'_C	$M_C^* = M_C \oplus K'_C$	Q_C	M_D	K'_D	$M_D^* = M_D \oplus K'_D$	Q_D
0	0	0	$ 0\rangle$	0	1	1	$ 1\rangle$	0	0	0	$ 0\rangle$
1	1	0	$ 0\rangle$	1	1	0	$ 0\rangle$	0	0	0	$ 0\rangle$
0	0	0	$ 0\rangle$	1	0	1	$ 1\rangle$	1	1	0	$ 0\rangle$
1	0	1	$ 1\rangle$	0	0	0	$ 0\rangle$	1	0	1	$ 1\rangle$
1	1	0	$ 0\rangle$	1	0	1	$ 1\rangle$	1	1	0	$ 0\rangle$
0	1	1	$ 1\rangle$	1	1	0	$ 0\rangle$	0	0	0	$ 0\rangle$
0	0	0	$ 0\rangle$	0	1	1	$ 1\rangle$	1	1	0	$ 0\rangle$
1	1	0	$ 0\rangle$	0	0	0	$ 0\rangle$	0	1	1	$ 1\rangle$
0	1	1	$ 1\rangle$	0	1	1	$ 1\rangle$	0	0	0	$ 0\rangle$
0	1	1	$ 1\rangle$	0	0	0	$ 0\rangle$	1	1	0	$ 0\rangle$
1	1	0	$ 0\rangle$	1	0	1	$ 1\rangle$	1	1	0	$ 0\rangle$
1	0	1	$ 1\rangle$	1	0	1	$ 1\rangle$	1	0	1	$ 1\rangle$
1	0	1	$ 1\rangle$	1	0	1	$ 1\rangle$	1	1	0	$ 0\rangle$
1	1	0	$ 0\rangle$	1	0	1	$ 1\rangle$	0	0	0	$ 0\rangle$
0	0	0	$ 0\rangle$	0	1	1	$ 1\rangle$	1	0	1	$ 1\rangle$
1	1	0	$ 0\rangle$	1	1	0	$ 0\rangle$	1	0	1	$ 1\rangle$

Finally, Bob, Charlie, and Dave can calculate the final shared key $K = K'_B \oplus K'_C \oplus K'_D$. Among them, the lengths of K'_B , K'_C , and K'_D are n bits. The reason is that in Step (4), $3n$ bit particles containing private key information in Cases 1–7 are used for eavesdropping detection, and only n bit particles containing private key information in Case 8 are used for key agreement.

It should be noted that due to the multiple transmissions of particles in the quantum channel, Alice, Bob, Charlie, and Dave need to install wavelength quantum filters and photon number separators to avoid Trojan horse attacks [29], [30].

III. ANALYSIS AND DISCUSSION

A. CORRECTNESS ANALYSIS

In this section, we demonstrate a specific example of the CASQKA protocol to analyze its correctness. For clarity, we leave out the steps of eavesdropping detection and inserting decoy particles. Assuming that $n = 4$, Bob, Charlie, and Dave secretly share the identity information sequences $R_B = \{1, 0, 0, 1\}$, $R_C = \{0, 1, 0, 1\}$, and $R_D = \{1, 1, 0, 0\}$ with Alice in advance. Bob, Charlie, and Dave randomly generate private keys: $K_B = \{1, 0, 0, 0, 0, 1, 1, 1, 0, 1, 1, 1, 1, 0, 1, 0\}$, $K_C = \{1, 0, 0, 1, 0, 1, 0, 1, 0, 1, 0, 0, 1, 1, 0\}$, and $K_D = \{1, 0, 0, 0, 0, 1, 1, 1, 1, 1, 0, 0, 0, 0, 1\}$.

Step 1: Bob, Charlie, and Dave prepare $r_B = |1\rangle|0\rangle|0\rangle|1\rangle$, $r_C = |0\rangle|1\rangle|0\rangle|1\rangle$, and $r_D = |1\rangle|1\rangle|0\rangle|0\rangle$, respectively. Then, Bob (Charlie and Dave) chooses $\Pi_4^B = \begin{pmatrix} 1234 \\ 3421 \end{pmatrix}$ ($\Pi_4^C = \begin{pmatrix} 1234 \\ 4312 \end{pmatrix}$, $\Pi_4^D = \begin{pmatrix} 1234 \\ 2314 \end{pmatrix}$) to rearrange $r_B(r_C, r_D)$ to obtain $r_B^* = |0\rangle|1\rangle|0\rangle|1\rangle$ ($r_C^* = |1\rangle|0\rangle|0\rangle|1\rangle$, $r_D^* = |1\rangle|0\rangle|1\rangle|0\rangle$), and chooses Π_{16}^B (Π_{16}^C, Π_{16}^D)

$$\Pi_{16}^B = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 & 12 & 13 & 14 & 15 & 16 \\ 3 & 1 & 2 & 4 & 6 & 7 & 5 & 8 & 12 & 10 & 11 & 9 & 16 & 13 & 14 & 15 \end{pmatrix} \quad (2)$$

$$\Pi_{16}^C = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 & 12 & 13 & 14 & 15 & 16 \\ 4 & 1 & 2 & 3 & 7 & 6 & 8 & 5 & 10 & 11 & 9 & 12 & 16 & 13 & 15 & 14 \end{pmatrix} \quad (3)$$

$$\Pi_{16}^D = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 & 12 & 13 & 14 & 15 & 16 \\ 2 & 4 & 1 & 3 & 7 & 5 & 6 & 8 & 11 & 9 & 10 & 12 & 16 & 14 & 15 & 13 \end{pmatrix} \quad (4)$$

to rearrange $K_B(K_C, K_D)$ to obtain $K_B^* = \{0, 1, 0, 0, 1, 1, 0, 1, 1, 1, 0, 0, 1, 0, 1\}$, $K_C^* = \{1, 1, 0, 0, 0, 1, 1, 0, 1, 0, 0, 0, 0, 1, 1\}$, and $K_D^* = \{0, 0, 1, 0, 1, 0, 1, 1, 0, 1, 1, 0, 1, 0, 0\}$.

Step 2: Alice prepares 32 cluster states $|\Phi\rangle_{ABCD}$ to form a cluster state sequence and divides it into sequences S_A , S_B , S_C , and S_D . Then, Alice sends sequences S_B , S_C , and S_D to Bob, Charlie, and Dave, respectively, keeping sequence S_A to herself.

Step 3: Bob, Charlie, and Dave randomly select 16 particles to perform CTRL operation, and Z-based measurements are performed on the remaining 16 particles. The measured results are recorded as M_B , M_C , and M_D , as shown in Table 3. From this table, we can also see the specific results of coding sequences M_B^* , M_C^* , and M_D^* and new particle sequences Q_B , Q_C , and Q_D . Then, Bob (Charlie and Dave) sends the new sequence $S_B^*(S_C^*, S_D^*)$ obtained by random combination to Alice.

Step 4: After receiving S_B^* , S_C^* , and S_D^* , Alice restores the sequences r_B^* , r_C^* , and r_D^* according to the published particle position information and performs Z-based measurement on them to obtain $r_B^* = |0\rangle|1\rangle|0\rangle|1\rangle$, $r_C^* = |1\rangle|0\rangle|0\rangle|1\rangle$, and $r_D^* = |1\rangle|0\rangle|1\rangle|0\rangle$ and then decodes them into binary sequences $R_B^* = \{0, 1, 0, 1\}$, $R_C^* = \{1, 0, 0, 1\}$, and $R_D^* = \{1, 0, 1, 0\}$. At the same time, the particles retained in Case 8 of Table 2 are recorded as sequences $Q'_B = |1\rangle|0\rangle|0\rangle|0\rangle$, $Q'_C = |1\rangle|1\rangle|1\rangle|0\rangle$, and $Q'_D = |0\rangle|0\rangle|1\rangle|1\rangle$ (corresponding to the last four rows of Q_B , Q_C , and Q_D in Table 3).

TABLE 4. Process of Obtaining the Key

M'_A	M'_B	M'_C	M'_D	$M_B^{*'} = M'_B \oplus K_B^{*'}$	$M_C^{*'} = M'_C \oplus K_C^{*'}$	$M_D^{*'} = M'_D \oplus K_D^{*'}$	$K_B^{*'}$	$K_C^{*'}$	$K_D^{*'}$
1	1	1	1	1	1	0	0	0	1
0	1	1	0	0	1	0	1	0	0
1	0	0	1	0	1	1	0	1	0
1	1	1	1	0	0	1	1	1	0

Step 5: Once the eavesdropping detection is successfully completed, Alice obtains $R'_B = \{1, 0, 0, 1\}$ ($R'_C = \{0, 1, 0, 1\}$, $R'_D = \{1, 1, 0, 0\}$) according to the permutation operators $\Pi_4^B(\Pi_4^C, \Pi_4^D)$ and $R_B^{*'}(R_C^{*'}, R_D^{*'})$. The comparison finds that $R'_B = R_B = \{1, 0, 0, 1\}$, $R'_C = R_C = \{0, 1, 0, 1\}$, and $R'_D = R_D = \{1, 1, 0, 0\}$ successfully authenticate the identity of Bob (Charlie and Dave).

Step 6: Bob (Charlie and Dave) restores $Q'_C = |1\rangle|1\rangle|1\rangle|0\rangle$ and $Q'_D = |0\rangle|0\rangle|1\rangle|1\rangle$ ($Q'_B = |1\rangle|0\rangle|0\rangle|0\rangle$ and $Q'_D = |0\rangle|0\rangle|1\rangle|1\rangle$, $Q'_B = |1\rangle|0\rangle|0\rangle|0\rangle$ and $Q'_C = |1\rangle|1\rangle|1\rangle|0\rangle$).

Step 7: Alice performs Z-based measurement on the sequence S_A , decodes the binary sequence M_A , and publishes it. Taking Bob as an example, Bob performs Z-based measurements on Q'_C and Q'_D and decodes them to obtain binary sequences $M_C^{*'} = \{1, 1, 1, 0\}$ and $M_D^{*'} = \{0, 0, 1, 1\}$. According to (1), $M'_A = \{1, 0, 1, 1\}$ and $M'_B = \{1, 1, 0, 1\}$, we can see $M'_C = \{1, 1, 0, 1\}$ and $M'_D = \{1, 0, 1, 1\}$. According to Table 4, Bob can get $K_C^{*'} = \{0, 0, 1, 1\}$ and $K_D^{*'} = \{1, 0, 0, 0\}$, and then, according to the permutation operators Π_{16}^C and Π_{16}^D , Bob can get $K'_C = \{0, 1, 1, 0\}$ and $K'_D = \{0, 0, 0, 1\}$. Similarly, Charlie can get $K'_B = \{1, 0, 1, 0\}$ and $K'_D = \{0, 0, 0, 1\}$; Dave can get $K'_B = \{1, 0, 1, 0\}$ and $K'_C = \{0, 1, 1, 0\}$. Finally, Bob, Charlie, and Dave can calculate the final shared key $K = K'_B \oplus K'_C \oplus K'_D = \{1, 0, 1, 0\} \oplus \{0, 1, 1, 0\} \oplus \{0, 0, 0, 1\} = \{1, 1, 0, 1\}$.

In summary, the protocol in this article is shown to be correct in this example.

B. SECURITY ANALYSIS

A secure SQKA protocol must be able to resist insider and outsider attacks. Since the protocol also carries out the process of identity authentication before the participants negotiate the shared key, when conducting security analysis, we must first consider whether the identity authentication will be attacked and then consider the key agreement part. Specific analysis is presented as follows.

1) INSIDER ATTACK

In this CASQKA protocol, Bob, Charlie, and Dave negotiate the shared key with the help of the trusted controller Alice. Bob, Charlie, and Dave contribute equally to the ultimately generated key, and no participant can determine the generation of the shared key independently.

It is assumed that Bob attempts to communicate with Alice by impersonating Charlie (Dave). Since Bob does not know the identity information sequence $R_C(R_D)$ of Charlie

(Dave), he can only randomly prepare the particle sequence $r'_C = r'_C{}^1 r'_C{}^2, \dots, r'_C{}^n$ ($r'_D = r'_D{}^1 r'_D{}^2, \dots, r'_D{}^n$) in Step (1). However, in Step (5), when Alice authenticates Charlie (Dave), the obtained $R'_C(R'_D)$ is obviously independent of the identity information sequence of Charlie (Dave). According to the preparation rules in Step (1), when the length of the identity information sequence is n , the probability of success of Bob impersonating Charlie (Dave) is $(1/2)^n$. When n is infinite, the probability of Bob's camouflage success approaches 0. Similarly, Charlie and Dave cannot pretend to be the other two participants.

Assuming that Bob is a dishonest participant, he attempts to determine the shared key $K = K'_B \oplus K'_C \oplus K'_D$ alone. Then, Bob needs to get K_C and K_D before sending the sequence S_B^* containing his private key information. However, in this protocol, the trusted controller Alice will not disclose any information about K_C and K_D before receiving the sequence S_B^* sent by Bob. In addition, only after Charlie and Dave receive the sequence Q_B^* , they announced the permutation operations Π_{4n}^C and Π_{4n}^D for the private key. Therefore, Bob cannot independently decide the creation of the shared key. Similarly, Charlie and Dave cannot generate the shared keys on their own.

Assuming that Bob and Charlie are two dishonest participants, they exchange their private key information in advance and try to jointly determine the shared key. Then, Bob and Charlie must first get the sequence containing Dave's private key information. However, Alice is a trusted controller in this protocol. Before receiving S_B^* and S_C^* from Bob and Charlie, Alice will not disclose any information about K_D . And Dave publishes the permutation operation Π_{4n}^D only after receiving the sequences $Q_B^{*'}$ and $Q_C^{*'}$ containing the private key information of Bob and Charlie. Therefore, Bob and Charlie cannot jointly determine the generation of the final shared key. Similarly, Bob and Dave (Charlie and Dave) cannot jointly determine the generation of shared keys.

2) OUTSIDER ATTACK

Although this protocol transmits information three times in the quantum channel, the particle sequence transmitted in Step (2) does not contain any secret information. Only sequences S_B^* , S_C^* , and S_D^* transmitted in Step (3) and sequences $Q_B^{*'}$, $Q_C^{*'}$, and $Q_D^{*'}$ transmitted in Step (6) contain useful information. If an attacker Eve wants to steal the final shared key, she can only achieve her goal by performing the following attacks on Steps (3) and (6).

a) *Attacks on identity authentication*

- 1) *Description of the attack method:* The attacker Eve obtains the secret information by impersonating the identity of one of the participants to communicate with other participants.
- 2) *Analysis of the protocol's defense mechanism:* It is assumed that the attacker Eve impersonates the identity of Bob (Charlie and Dave) to communicate with Alice. Since Eve does not know the identity information sequence $R_B(R_C, R_D)$ of Bob (Charlie and Dave), she can only randomly prepare the particle sequence $r'_B = r_B^1 r_B^2, \dots, r_B^m (r'_C = r_C^1 r_C^2, \dots, r_C^m, r'_D = r_D^1 r_D^2, \dots, r_D^m)$ in Step (1). However, in Step (5), when Alice authenticates Bob (Charlie and Dave), the obtained $R'_B(R'_C, R'_D)$ is obviously independent of the identity information sequence of Bob (Charlie and Dave). According to the preparation rules in Step (1), when the length of the identity information sequence is n , the probability of success of Eve impersonating Bob (Charlie and Dave) is $(1/2)^n$. When n is infinite, the probability of Eve's camouflage success approaches 0.

b) *Trojan horse attack*

- 1) *Description of the attack method:* Trojan horse attack is mainly divided into two forms: invisible photon eavesdropping (IPE) Trojan horse attack and delay-photon Trojan horse attack. The IPE Trojan horse attack means that the attacker sends invisible photons outside the normal band of the system into the QKA system, and the photons return after interacting with the components of the system. The attacker obtains sensitive information by measuring and analyzing the reflected photons. The delay-photon Trojan horse attack means that the attacker sends strong pulse photons with special wavelengths to the receiver of the QKA system. The photons interact with the components inside the device, and some of the photons are delayed after being reflected and scattered. The attacker captures and analyzes these delayed photons to infer the working state or secret information of the device.
- 2) *Analysis of the protocol's defense mechanism:* In this protocol, the particles are transmitted three times in the channel. To resist two Trojan horse attacks, this protocol introduces two devices: wavelength quantum filter (which restricts the wavelengths of photons that can pass through to prevent the IPE Trojan horse attack) and photon number separator (which separates a large number of photons contained in the strong pulsed light to prevent the delay-photon Trojan horse attack) [7]. These two devices blocked the Trojan horse attack.

c) *Intercept-resend attack*

- 1) *Description of the attack method:* The attacker Eve first intercepts the quantum sequence sent by the sender,

then generates a new forged sequence, and sends it to the receiver. After receiving the forged sequence, the receiver will encode the particles containing their identity information and private key information into the forged sequence to form a new sequence, and feedback the new sequence back. The purpose of Eve is to obtain the secret information of the receiver by measuring the new sequence fed back by the receiver. Since the forged sequence is prepared by Eve, she knows the initial state of the forged sequence. By measuring the new sequence reflected and comparing the measurement results with the initial state, Eve can infer the secret information of the receiver.

- 2) *Analysis of the protocol's defense mechanism:* Take Step (3) as an example of Eve intercepting sequence $S_B^*(S_C^*, S_D^*)$ sent by Bob (Charlie and Dave) to Alice. After intercepting the sequence $S_B^*(S_C^*, S_D^*)$, Eve forges a sequence of the same length as $S_B^*(S_C^*, S_D^*)$ and sends it to Alice. Since Eve does not know which particles perform the CTRL operation by Bob (Charlie and Dave), her malicious operation will be found when Alice performs eavesdropping detection in Step (4).

Similarly, in Step (6), Eve intercepts sequences Q_B^* , Q_C^* , and Q_D^* sent by Alice. As Eve is unaware of the decoy particles' positions, her attack behavior will be detected during the security check in Step (8). Therefore, the protocol can resist the intercept-resend attack.

d) *Measure-resend attack*

- 1) *Description of the attack method:* After intercepting the quantum sequence, the attacker Eve measures the particles in the sequence to try to obtain the quantum state information, then regenerates the corresponding particles according to the measurement results, and sends them to the receiver. This attack aims to use the quantum state information obtained by measurement to crack the key or obtain the secret information.
- 2) *Analysis of the protocol's defense mechanism:* Take Eve's measure-resend attack on the sequence $S_B^*(S_C^*, S_D^*)$ sent by Bob (Charlie and Dave) to Alice as an example. After Eve intercepts the sequence $S_B^*(S_C^*, S_D^*)$, Z-based measurement is performed on it. Based on the measurement results, the particle sequence with the same state is prepared and sent to Alice. Similarly, since Eve is unaware of which particles are performed CTRL operation, the existence of attacker Eve will be found when Alice performs eavesdropping detection in Step (4). Next, consider the sequences Q_B^* , Q_C^* , and Q_D^* sent by Alice intercepted by Eve in Step (6) as an example; because Eve is unaware of the position of the decoy particles, she cannot pass the eavesdropping detection in Step (8). Therefore, the protocol can resist measure-resend attack.

TABLE 5. Comparison Between the CASQKA Protocol in This Article and the Existing QKA Protocols

Protocol	Quantum resource	Identity authentication function	Semi quantum properties	Qubit efficiency
Ma et al.'s protocol ^[17]	five-qubit genuinely entangled states	√	×	7.7%
He et al.'s protocol ^[18]	Bell states	√	×	16.67%
He et al.'s protocol ^[19]	single-particle states and four-particle GHZ states	√	×	53.33%
Wang et al.'s protocol ^[21]	single-particle states	√	×	16.67%
He et al.'s protocol ^[23]	G-like states	×	√	4.5%
He et al.'s protocol ^[25]	Cluster states	×	√	1.6%
Zhou et al.'s protocol ^[31]	Cluster states	×	√	2.08%
Yan et al.'s protocol ^[34]	Bell states	×	√	6.7%
Our protocol	Cluster states	√	√	2%

e) Entangle-measure attack

- 1) *Description of the attack method:* Entangle-measure attack utilizes the characteristics of quantum entanglement, that is, the quantum states of two or more quantum systems can be correlated together in some way, so that information about another system can be obtained by measuring one system. In this attack, Eve first introduces the auxiliary state particles to entangle with the target particles of the communicator, and then, by measuring the auxiliary state particles, the secret information encoded on the target particles can be obtained.
- 2) *Analysis of the protocol's defense mechanism:* Suppose that Eve is prepared to use the auxiliary state particle $|E\rangle$ to perform an entanglement operation U on $|\Phi\rangle_{ABCD}$ to form a five-qubit entangled state

$$U|\Phi\rangle_{ABCD}|E\rangle = \frac{1}{2} \begin{pmatrix} |0000\rangle_{ABCD} (|e_{0,0}\rangle + |e_{1,0}\rangle + |e_{2,0}\rangle + |e_{3,0}\rangle) \\ + |0110\rangle_{ABCD} (|e_{0,1}\rangle + |e_{1,1}\rangle + |e_{2,1}\rangle + |e_{3,1}\rangle) \\ + |1001\rangle_{ABCD} (|e_{0,2}\rangle + |e_{1,2}\rangle + |e_{2,2}\rangle + |e_{3,2}\rangle) \\ + |1111\rangle_{ABCD} (|e_{0,3}\rangle + |e_{1,3}\rangle + |e_{2,3}\rangle + |e_{3,3}\rangle) \end{pmatrix} \quad (5)$$

where $|e_{0,0}\rangle, |e_{1,0}\rangle \dots |e_{3,3}\rangle$ are the pure states uniquely determined by U .

Bob, Charlie, and Dave can only get two results $|0\rangle$ or $|1\rangle$ by measuring their particles with Z-based measurement. The

hybrid system of Alice, Bob, Charlie, Dave, and Eve is

$$\rho_{ABCDE} = \frac{1}{4} \begin{pmatrix} |0\rangle_A |0\rangle \otimes |0\rangle_B |0\rangle \otimes |0\rangle_C |0\rangle \otimes |0\rangle_D |0\rangle \\ (|e_{0,0}\rangle \langle e_{0,0}| + |e_{1,0}\rangle \langle e_{1,0}| + |e_{2,0}\rangle \langle e_{2,0}| + |e_{3,0}\rangle \langle e_{3,0}|) \\ + |0\rangle_A |0\rangle \otimes |1\rangle_B |1\rangle \otimes |1\rangle_C |1\rangle \otimes |0\rangle_D |0\rangle \\ (|e_{0,1}\rangle \langle e_{0,1}| + |e_{1,1}\rangle \langle e_{1,1}| + |e_{2,1}\rangle \langle e_{2,1}| + |e_{3,1}\rangle \langle e_{3,1}|) \\ + |1\rangle_A |1\rangle \otimes |0\rangle_B |0\rangle \otimes |0\rangle_C |0\rangle \otimes |1\rangle_D |1\rangle \\ (|e_{0,2}\rangle \langle e_{0,2}| + |e_{1,2}\rangle \langle e_{1,2}| + |e_{2,2}\rangle \langle e_{2,2}| + |e_{3,2}\rangle \langle e_{3,2}|) \\ + |1\rangle_A |1\rangle \otimes |1\rangle_B |1\rangle \otimes |1\rangle_C |1\rangle \otimes |1\rangle_D |1\rangle \\ (|e_{0,3}\rangle \langle e_{0,3}| + |e_{1,3}\rangle \langle e_{1,3}| + |e_{2,3}\rangle \langle e_{2,3}| + |e_{3,3}\rangle \langle e_{3,3}|) \end{pmatrix} \quad (6)$$

According to [31] and [32], it can be inferred that the probability of $|0\rangle$ obtained by Alice (Bob, Charlie, and Dave) after performing Z-based measurement is $P_{A_0} (P_{B_0}, P_{C_0}, P_{D_0})$; the probability of $|1\rangle$ is $P_{A_1} (P_{B_1}, P_{C_1}, P_{D_1})$

$$P_{A_0} = P_{D_0} = \frac{1}{4} \begin{pmatrix} \langle e_{0,0} | e_{0,0} \rangle + \langle e_{1,0} | e_{1,0} \rangle + \langle e_{2,0} | e_{2,0} \rangle \\ + \langle e_{3,0} | e_{3,0} \rangle + \langle e_{0,1} | e_{0,1} \rangle + \langle e_{1,1} | e_{1,1} \rangle \\ + \langle e_{2,1} | e_{2,1} \rangle + \langle e_{3,1} | e_{3,1} \rangle \end{pmatrix} \quad (7)$$

$$P_{A_1} = P_{D_1} = \frac{1}{4} \begin{pmatrix} \langle e_{0,2} | e_{0,2} \rangle + \langle e_{1,2} | e_{1,2} \rangle + \langle e_{2,2} | e_{2,2} \rangle \\ + \langle e_{3,2} | e_{3,2} \rangle + \langle e_{0,3} | e_{0,3} \rangle + \langle e_{1,3} | e_{1,3} \rangle \\ + \langle e_{2,3} | e_{2,3} \rangle + \langle e_{3,3} | e_{3,3} \rangle \end{pmatrix} \quad (8)$$

$$P_{B_0} = P_{C_0} = \frac{1}{4} \begin{pmatrix} \langle e_{0,0} | e_{0,0} \rangle + \langle e_{1,0} | e_{1,0} \rangle + \langle e_{2,0} | e_{2,0} \rangle \\ + \langle e_{3,0} | e_{3,0} \rangle + \langle e_{0,2} | e_{0,2} \rangle + \langle e_{1,2} | e_{1,2} \rangle \\ + \langle e_{2,2} | e_{2,2} \rangle + \langle e_{3,2} | e_{3,2} \rangle \end{pmatrix} \quad (9)$$

$$P_{B_1} = P_{C_1} = \frac{1}{4} \begin{pmatrix} \langle e_{0,1} | e_{0,1} \rangle + \langle e_{1,1} | e_{1,1} \rangle + \langle e_{2,1} | e_{2,1} \rangle \\ + \langle e_{3,1} | e_{3,1} \rangle + \langle e_{0,3} | e_{0,3} \rangle + \langle e_{1,3} | e_{1,3} \rangle \\ + \langle e_{2,3} | e_{2,3} \rangle + \langle e_{3,3} | e_{3,3} \rangle \end{pmatrix}. \quad (10)$$

The Shannon entropy of the Dave system (D) is $H(D) = h(P_{D_0}, P_{D_1}) = 1$ in the absence of an attacker. According to the measurement correlation of the cluster state $|\Phi\rangle_{ABCD}$, the trusted controller Alice (A) can infer the state of the corresponding particle of Dave according to the measurement results of the particles in her hands; that is to say, conditional entropy $H(D|A) = 0$, and mutual information $I(A : D) = H(D) - H(D|A) = 1$.

The mutual information between Alice (A) and attacker Eve (E) is denoted as $I(A : E)$. To prevent Eve's attack behavior from being discovered, the measurement result obtained by Alice must be $|\Phi\rangle_{ABCD}$ when Bob, Charlie, and Dave perform CTRL operations on the corresponding particles. Therefore, (6) should satisfy

$$\begin{aligned} & (|e_{0,0}\rangle\langle e_{0,0}| + |e_{1,0}\rangle\langle e_{1,0}| + |e_{2,0}\rangle\langle e_{2,0}| + |e_{3,0}\rangle\langle e_{3,0}|) \\ &= (|e_{0,1}\rangle\langle e_{0,1}| + |e_{1,1}\rangle\langle e_{1,1}| + |e_{2,1}\rangle\langle e_{2,1}| + |e_{3,1}\rangle\langle e_{3,1}|) \\ &= (|e_{0,2}\rangle\langle e_{0,2}| + |e_{1,2}\rangle\langle e_{1,2}| + |e_{2,2}\rangle\langle e_{2,2}| + |e_{3,2}\rangle\langle e_{3,2}|) \\ &= (|e_{0,3}\rangle\langle e_{0,3}| + |e_{1,3}\rangle\langle e_{1,3}| + |e_{2,3}\rangle\langle e_{2,3}| + |e_{3,3}\rangle\langle e_{3,3}|). \end{aligned} \quad (11)$$

The mutual information between Alice and Dave is $I(A : E) = 0$. Since $I(A : D) = 1 > I(A : E) = 0$, the attacker Eve is unable to acquire any valuable information. Thus, this protocol can withstand an entangle-measure attack.

C. ANALOG SIMULATION

Due to the fact that real quantum computer devices are extremely scarce and face numerous technical limitations, we choose to utilize the simulation environment to verify the protocol. The design of quantum circuit is crucial for the simulation of the protocol. The quantum circuit diagram of the CASQKA protocol is constructed by IBM Qiskit, which is divided into eight cases according to Table 2, as shown in Figs. 4–11. In quantum circuits, $q[0]$, $q[1]$, $q[2]$, and $q[3]$ represent four particles, respectively, which generate cluster states through quantum logic H gates, CZ gates and CNOT gates. On the right-hand side of the first dividing line, the operations on the three quantum circuits $q[1]$, $q[2]$, and $q[3]$ are performed by Bob, Charlie, and Dave, respectively. On the right-hand side of the second dividing line, all operations on the quantum circuit are performed by Alice.

The security of the protocol mainly relies on the entanglement properties of the cluster states. As shown in Table 2,

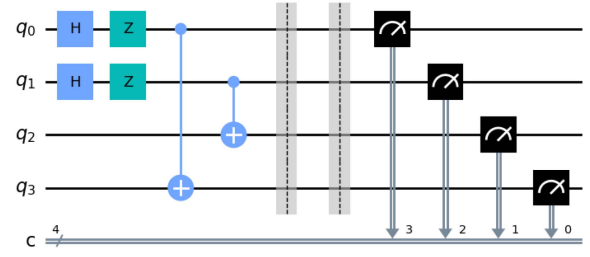


FIGURE 4. Quantum circuit of Case 1. Bob, Charlie, and Dave all perform CTRL operation, and Alice performs cluster joint measurement.

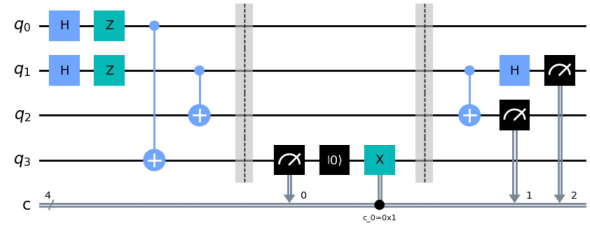


FIGURE 5. Quantum circuit of Case 2. Bob and Charlie perform CTRL operation, Dave performs SIFT operation, and Alice performs Bell measurement on Bob and Charlie's particles.

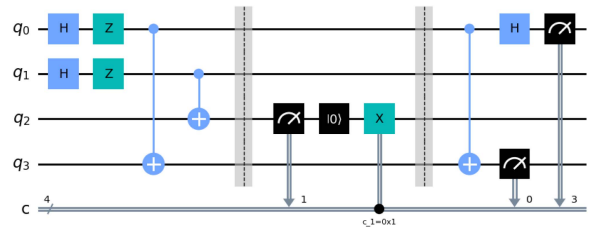


FIGURE 6. Quantum circuit of Case 3. Bob and Dave perform CTRL operation, Charlie performs SIFT operation, and Alice performs Bell measurement on the particles in her hand and the particles of Dave.

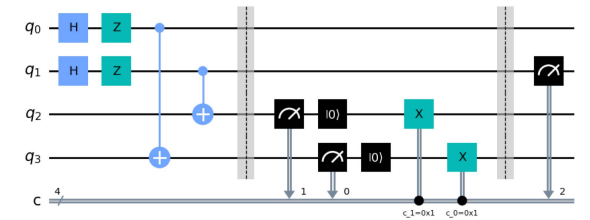


FIGURE 7. Quantum circuit of Case 4. Bob performs CTRL operation, Charlie and Dave perform SIFT operation, and Alice performs Z-based measurement on Bob's particles.

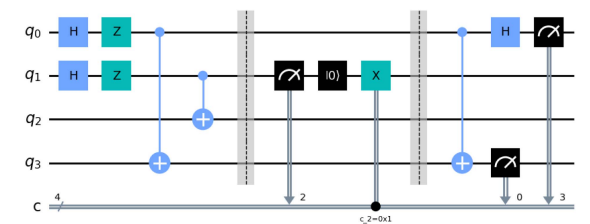


FIGURE 8. Quantum circuit of Case 5. Bob performs SIFT operation, Charlie and Dave perform CTRL operation, and Alice performs Bell measurement on the particles in her hand and the particles of Dave.

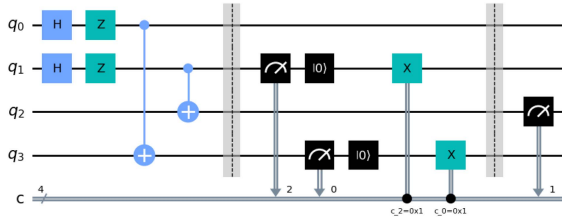


FIGURE 9. Quantum circuit of Case 6. Bob and Dave perform SIFT operation, Charlie performs CTRL operation, and Alice performs Z-based measurement on Charlie's particles.

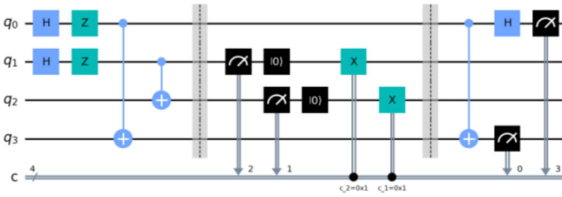


FIGURE 10. Quantum circuit of Case 7. Bob and Charlie perform SIFT operation, Dave performs CTRL operation, and Alice performs Bell measurement on the particles in her hand and the particles of Dave.

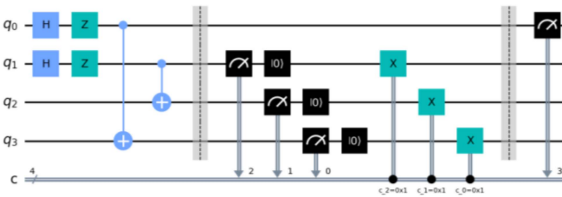


FIGURE 11. Quantum circuit of Case 8. Bob, Charlie, and Dave all perform SIFT operations, and Alice performs Z-based measurements on her particles.

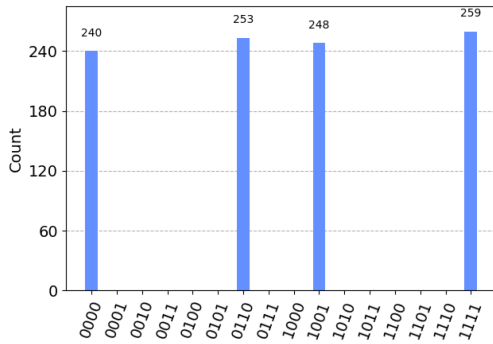


FIGURE 12. Results of running Fig. 11.

the security checks for Cases 1–7 are valid. For the particles in Case 8 used for key agreements, we can see whether these particles are still entangled through simulation. Fig. 11 shows the circuit diagram of Case 8 after adding Alice to perform Z-based measurement on the particles, aiming to verify whether the particles retained in Case 8 are still entangled with the particles at the corresponding position in Alice's hand. The running result of Fig. 11 is shown in Fig. 12. According to Fig. 12, the measurement results of the particles are evenly distributed in the four cases of $|0000\rangle$, $|0110\rangle$, $|1001\rangle$, and $|1111\rangle$, which conforms to the description of the

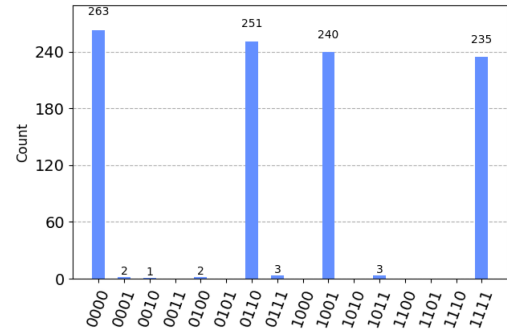


FIGURE 13. Results of Fig. 11 under the noisy environment with a noise parameter of 0.01.

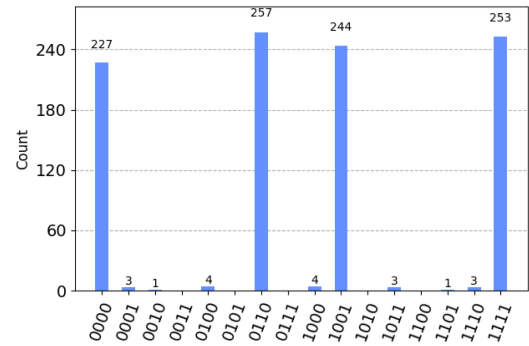


FIGURE 14. Results of Fig. 11 under the noisy environment with a noise parameter of 0.02.

cluster states in (1). This shows that the particles used for key agreements are still entangled. Therefore, it can be confirmed that the key agreement process of the protocol is secure.

In addition, the following two types of quantum noise are introduced into the quantum circuit of Case 8 to simulate the state of the particles used to negotiate the key in noisy environments.

- I) *Single-qubit noise*: It is usually described by the amplitude damping noise model, which is suitable for all single-qubit gates, such as X gates, H gates, and RY gates. Amplitude damping noise is mainly the simulation of qubit energy dissipation. Under the influence of this noise, when the qubit is in the excited state $|1\rangle$, it will decay to the ground state $|0\rangle$ with a certain probability, which will change the state of the qubit.
- II) *Two-qubit noise*: It is usually described by a depolarization noise model, which is suitable for all two-qubit gates, such as CX gates and CZ gates. Depolarization noise will make the quantum bits originally in a specific entangled state gradually lose their entanglement characteristics with a certain probability, and become a uniform mixing of various states.

The noise parameters are set to 0.01, 0.02, 0.03, and 0.04, respectively. That is, the probability of noise in each quantum gate operation is 1%, 2%, 3%, and 4%. The results obtained are shown in Figs. 13–16. According to Figs. 13–16, the measurement results of the particles are still mainly evenly

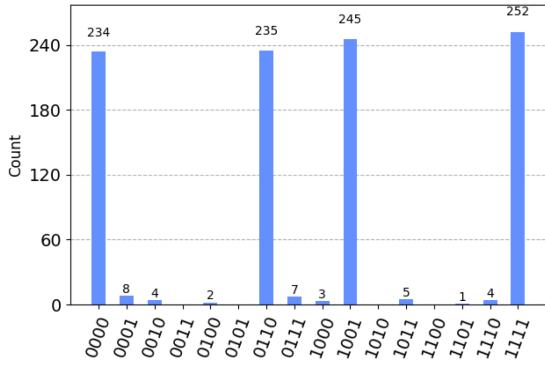


FIGURE 15. Results of Fig. 11 under the noisy environment with a noise parameter of 0.03.

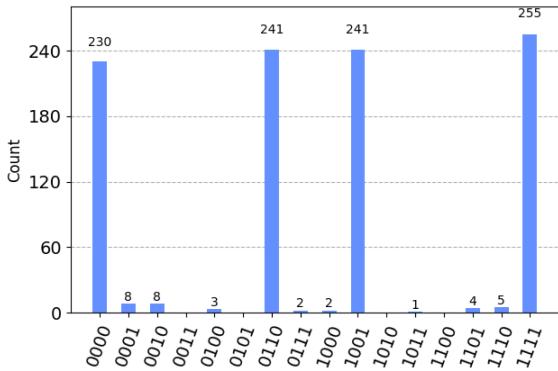


FIGURE 16. The results of Fig. 11 under the noisy environment with a noise parameter of 0.04.

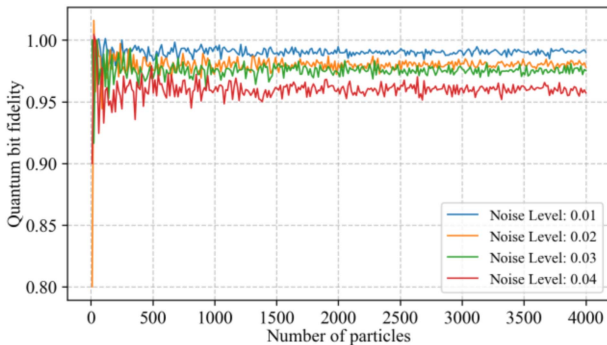


FIGURE 17. Quantum bit fidelity in noisy environments with different noise parameters.

distributed in the four cases of $|0000\rangle$, $|0110\rangle$, $|1001\rangle$, and $|1111\rangle$, which correspond to the description of the cluster states in (1).

Fig. 17 describes the similarity between the actual and ideal states of the qubit in the noise environment with parameters of 0.01, 0.02, 0.03 and 0.04, that is, the fidelity of the qubit. The value range of fidelity is between 0 and 1, 1 means that the actual state is exactly the same as the ideal state, that is, there is no error. 0 means that the actual state is completely different from the ideal state. It can be seen from Fig. 17 that as the number of particles increases, the fidelity of the qubit gradually tends to be stable. Under the influence

of the noise of the four parameters, the fidelity of the qubit can still be maintained above 0.95, which further indicates that the protocol has strong robustness and stability in the noise environment, which can effectively suppress the noise interference and ensure the accuracy of quantum information processing.

D. PERFORMANCE ANALYSIS

The performance of QKA protocols is usually described by Cabello qubit efficiency [33]. The Cabello qubit efficiency is defined as $\eta = c/(q + b)$, where q is the total number of qubits, b denotes the number of exchanged classical qubits, and c represents the bit length of the final shared key.

The length of the shared key is $c = n$; the number of cluster states prepared by Alice is $8n$; the number of identity information particles is $3n$, that is $q = 8n \times 4 + n \times 3 = 35n$. The bit numbers of the permutation operators $\Pi_n^B(\Pi_n^C, \Pi_n^D)$ and $\Pi_{4n}^B(\Pi_{4n}^C, \Pi_{4n}^D)$ are n and $4n$, respectively, that is, $b = n \times 3 + 4n \times 3 = 15n$. Therefore, the quantum bit efficiency of this protocol is $\eta = n/(35n + 15n) = 2\%$. Table 5 shows the performance comparison between this protocol and other existing QKA protocols. According to Table 5, compared with the protocols [17], [18], [19], [21], the new protocol allows semiquantum parties to participate to reduce the requirements for participants' quantum capabilities and quantum devices. Compared with the protocols [23], [25], [31], [34], the new protocol has identity authentication function. Table 5 shows that, in comparison to the traditional QKA protocols, the efficiency of SQKA protocols is significantly reduced, and our protocol also adds new functions to SQKA. The more functions the protocol supports simultaneously, the more particles it consumes, leading to an inevitable decrease in qubit efficiency. However, the quantum bit efficiency of this protocol is 2%, and the efficiency of this SQKA protocol is acceptable.

IV. CONCLUSION

In WITMED, to ensure the confidentiality and integrity of the data generated by the doctors participating in the joint consultation when discussing the patient's condition and diagnosis and treatment plan, this article proposes a three-party CASQKA protocol based on the four-particle cluster state. With the help of the trusted controller Alice, who has full quantum capabilities (representing a hospital or a large institution with a strong quantum device), the three semiquantum parties Bob, Charlie, and Dave (representing the doctors participating in the meeting) perform identity authentication. After the authentication is passed, the shared key is negotiated fairly. The scheme not only has low requirements on the quantum devices of doctors participating in key agreements, but also prevents external attackers from stealing the shared key by counterfeiting the authorized doctor's identity. Finally, by comparing this protocol with the existing QKA protocol, it is found that the quantum bit efficiency of this protocol is acceptable under the condition of satisfying multiple functions. In summary, this agreement is more suitable

for patients with malignant tumors and other diseases that require multiple experts to conduct multidisciplinary joint consultations at the same time. Considering the need for more doctors to participate in consultations in practical applications, the design of an authenticated SQKA protocol that supports N participants using multiparticle entangled states will be one of the key topics in our future research.

REFERENCES

- [1] C. H. Bennett and G. Brassard, "Quantum cryptography: Public key distribution and coin tossing," *Theor. Comput. Sci.*, vol. 560, pp. 7–11, 2014, doi: [10.1016/j.tcs.2014.05.025](https://doi.org/10.1016/j.tcs.2014.05.025).
- [2] G. D. Kang, J. Liu, T. Zhang, Q.-P. Zhou, and M.-F. Fang, "Fully measurement-device-independent two-way quantum key distribution with finite single photon sources," *Quantum Inf. Process.*, vol. 23, 2024, Art. no. 211, doi: [10.1007/s11128-024-04419-5](https://doi.org/10.1007/s11128-024-04419-5).
- [3] N. Zhou, G. Zeng, and J. Xiong, "Quantum key agreement protocol," *Electron. Lett.*, vol. 40, no. 18, pp. 1149–1150, Sep. 2004, doi: [10.1049/el:20045183](https://doi.org/10.1049/el:20045183).
- [4] D. S. Shen, W. P. Ma, and L. Wang, "Two-party quantum key agreement with four-qubit cluster states," *Quantum Inf. Process.*, vol. 13, pp. 2313–2324, 2014, doi: [10.1007/s11128-014-0785-z](https://doi.org/10.1007/s11128-014-0785-z).
- [5] R. H. Shi and H. Zhong, "Multi-party quantum key agreement with Bell states and Bell measurements," *Quantum Inf. Process.*, vol. 12, pp. 921–932, 2013, doi: [10.1007/s11128-012-0443-2](https://doi.org/10.1007/s11128-012-0443-2).
- [6] B. Liu, F. Gao, W. Huang, and Q. Y. Wen, "Multiparty quantum key agreement with single particles," *Quantum Inf. Process.*, vol. 12, pp. 1797–1805, 2013, doi: [10.1007/s11128-012-0492-6](https://doi.org/10.1007/s11128-012-0492-6).
- [7] Y. F. He and W. P. Ma, "Quantum key agreement protocols with four-qubit cluster states," *Quantum Inf. Process.*, vol. 14, pp. 3483–3498, 2015, doi: [10.1007/s11128-015-1060-7](https://doi.org/10.1007/s11128-015-1060-7).
- [8] Z. Sun, J. Yu, and P. Wang, "Efficient multi-party quantum key agreement by cluster states," *Quantum Inf. Process.*, vol. 15, pp. 373–384, 2016, doi: [10.1007/s11128-015-1155-1](https://doi.org/10.1007/s11128-015-1155-1).
- [9] H. Yang, S. F. Lu, Q. Zhou, M. Wang, B. Y. Feng, and X. J. Zhou, "Efficient single-state multi-party quantum key agreement," *Quantum Inf. Process.*, vol. 23, no. 4, 2024, Art. no. 150, doi: [10.1007/s11128-024-04350-9](https://doi.org/10.1007/s11128-024-04350-9).
- [10] J. W. Liu, Q. L. Wang, Q. X. Mu, Y. C. Li, L. Cheng, and W. Q. Sun, "Advancing multi-party quantum key agreement protocols: The power of a two-layer approach," *Quantum Inf. Process.*, vol. 23, no. 3, 2024, Art. no. 97, doi: [10.1007/s11128-024-04302-3](https://doi.org/10.1007/s11128-024-04302-3).
- [11] Y. F. He and W. P. Ma, "Two quantum key agreement protocols immune to collective noise," *Int. J. Theor. Phys.*, vol. 56, pp. 328–338, 2017, doi: [10.1007/s10773-016-3165-x](https://doi.org/10.1007/s10773-016-3165-x).
- [12] L. Li, R. G. Zhou, and X. X. Zhang, "Three-party quantum key agreement protocol based on logical four-particle cluster state to resist collective noise," *Quantum Inf. Process.*, vol. 22, no. 12, 2023, Art. no. 453, doi: [10.1007/s11128-023-04206-8](https://doi.org/10.1007/s11128-023-04206-8).
- [13] S. X. Jiang, L. Fang, and X. J. Fang, "Two-party quantum key agreement with six-particle entangled states against collective noise," *Int. J. Theor. Phys.*, vol. 62, 2023, Art. no. 235, doi: [10.1007/s10773-023-05414-9](https://doi.org/10.1007/s10773-023-05414-9).
- [14] J. Tang, L. Shi, and J. H. Wei, "Controlled quantum key agreement based on maximally three-qubit entangled states," *Mod. Phys. Lett. B*, vol. 34, no. 18, 2020, Art. no. 2050201, doi: [10.1142/S0217984920502012](https://doi.org/10.1142/S0217984920502012).
- [15] D. Arindam and P. Anirban, "Collective attack free controlled quantum key agreement without quantum memory," *Phys. Scripta*, vol. 100, no. 3, 2025, Art. no. 035101, doi: [10.1088/1402-4896/adaa30](https://doi.org/10.1088/1402-4896/adaa30).
- [16] H. F. Zhu, C. N. Wang, and Z. X. Li, "Semi-honest three-party mutual authentication quantum key agreement protocol based on GHZ-like state," *Int. J. Theor. Phys.*, vol. 60, pp. 293–303, 2021, doi: [10.1007/s10773-020-04692-x](https://doi.org/10.1007/s10773-020-04692-x).
- [17] X. Ma, J. Hur, Z. Li, and H. Zhu, "Quantum mutual authentication key agreement scheme using five-qubit entanglement towards different realm architecture," *Int. J. Theor. Phys.*, vol. 60, no. 5, pp. 1933–1948, 2021, doi: [10.1007/s10773-021-04812-1](https://doi.org/10.1007/s10773-021-04812-1).
- [18] Y. F. He, Y. B. Pang, and M. Di, "Mutual authentication quantum key agreement protocol based on Bell states," *Quantum Inf. Process.*, vol. 21, no. 8, 2022, Art. no. 290, doi: [10.1007/s11128-022-03640-4](https://doi.org/10.1007/s11128-022-03640-4).
- [19] Y. F. He, Y. R. Yue, M. Di, and W. P. Ma, "Two-party mutual authentication quantum key agreement protocol," *Int. J. Theor. Phys.*, vol. 61, no. 5, 2022, Art. no. 145, doi: [10.1007/s10773-022-05125-7](https://doi.org/10.1007/s10773-022-05125-7).
- [20] Y. Xu, C. Wang, K. Cheng, and H. Zhu, "A novel three-party mutual authentication quantum key agreement protocol with GHZ states," *Int. J. Theor. Phys.*, vol. 61, no. 10, 2022, Art. no. 245, doi: [10.1007/s10773-022-05220-9](https://doi.org/10.1007/s10773-022-05220-9).
- [21] C. N. Wang, Q. Zhang, S. F. Liang, and H. F. Zhu, "Secure mutual authentication quantum key agreement scheme for two-party setting with key recycling," *Quantum Inf. Process.*, vol. 23, no. 4, 2024, Art. no. 139, doi: [10.1007/s11128-024-04356-3](https://doi.org/10.1007/s11128-024-04356-3).
- [22] C. Shukla, K. Thapliyal, and A. Pathak, "Semi-quantum communication: Protocols for key agreement, controlled secure direct communication and dialogue," *Quantum Inf. Process.*, vol. 16, no. 12, 2017, Art. no. 295, doi: [10.1007/s11128-017-1736-2](https://doi.org/10.1007/s11128-017-1736-2).
- [23] Y. F. He, Y. Pang, M. Di, Y. Yue, G. Li, and J. Liu, "A two-party semi-quantum key agreement protocol based on G-like states," *Chin. J. Lasers*, vol. 49, no. 13, 2022, Art. no. 1312001, doi: [10.3788/CJL202249.1312001](https://doi.org/10.3788/CJL202249.1312001).
- [24] T. J. Xu, Y. Chen, M. J. Geng, and T. Y. Ye, "Single-state multi-party semi quantum key agreement protocol based on multi-particle GHZ entangled states," *Quantum Inf. Process.*, vol. 21, no. 7, 2022, Art. no. 266, doi: [10.1007/s11128-022-03615-5](https://doi.org/10.1007/s11128-022-03615-5).
- [25] Y. F. He, Y. Pang, M. Di, Y. Yue, J. Liu, and G. Li, "Four-party semi-quantum key agreement protocol based on four-particle cluster states," *Acta Opt. Sin.*, vol. 43, no. 20, 2023, Art. no. 2027001, doi: [10.3788/AOS230828](https://doi.org/10.3788/AOS230828).
- [26] H. M. Yi, R. G. Zhou, and R. Q. Xu, "Semi-quantum key agreement protocol using W states," *Int. J. Theor. Phys.*, vol. 62, 2023, Art. no. 212, doi: [10.1007/s10773-023-05467-w](https://doi.org/10.1007/s10773-023-05467-w).
- [27] H. J. Briegel and R. Raussendorf, "Persistent entanglement in arrays of interacting particles," *Phys. Rev. Lett.*, vol. 86, no. 5, pp. 910–913, 2001, doi: [10.1103/PhysRevLett.86.910](https://doi.org/10.1103/PhysRevLett.86.910).
- [28] A. Cross, "The IBM Q experience and Qiskit open-source quantum computing software," *Bull. Amer. Phys. Soc.*, vol. 63, 2018, Art. no. L58.003. [Online]. Available: <https://meetings.aps.org/Meeting/MAR18/Session/L58.3>
- [29] Q. Y. Cai, "Eavesdropping on the two-way quantum communication protocols with invisible photons," *Phys. Lett. A*, vol. 351, nos. 1/2, pp. 23–25, 2006, doi: [10.1016/j.physleta.2005.10.050](https://doi.org/10.1016/j.physleta.2005.10.050).
- [30] F. G. Deng, X. H. Li, H. Y. Zhou, and Z. Zhang, "Improving the security of multiparty quantum secret sharing against Trojan horse attack," *Phys. Rev. A*, vol. 72, no. 4, 2005, Art. no. 044302, doi: [10.1103/PhysRevA.72.044302](https://doi.org/10.1103/PhysRevA.72.044302).
- [31] N. R. Zhou, K. N. Zhu, and Y. Q. Wang, "Three-party semi-quantum key agreement protocol," *Int. J. Theor. Phys.*, vol. 59, no. 3, pp. 663–676, 2020, doi: [10.1007/s10773-019-04288-0](https://doi.org/10.1007/s10773-019-04288-0).
- [32] L. Y. Chen, L. H. Gong, and N. R. Zhou, "Two semi-quantum key distribution protocols with G-like states," *Int. J. Theor. Phys.*, vol. 59, no. 6, pp. 1884–1896, 2020, doi: [10.1007/s10773-020-04456-7](https://doi.org/10.1007/s10773-020-04456-7).
- [33] A. Cabello, "Quantum key distribution in the Holevo limit," *Phys. Rev. Lett.*, vol. 85, no. 26, pp. 5635–5638, 2000, doi: [10.1103/PhysRevLett.85.5635](https://doi.org/10.1103/PhysRevLett.85.5635).
- [34] L. Yan, S. Zhang, Y. Chang, Z. Sheng, and Y. Sun, "Semi-quantum key agreement and private comparison protocols using Bell states," *Int. J. Theor. Phys.*, vol. 58, pp. 3852–3862, 2019, doi: [10.1007/s10773-019-04252-y](https://doi.org/10.1007/s10773-019-04252-y).